Theory

R is a powerful open-source language designed for statistical computing, data analysis, and graphical representation. Before writing any program, it is essential to set up the working directory so R knows where to read/write files. R also provides a rich set of operators and built-in functions for performing calculations.

1. Working Directory Functions:

- getwd() — Returns the current working directory path.
- setwd() — Sets a new working directory.
- list.files() — Lists all files in the current working directory.

2. Arithmetic Operators:

- + Addition | - Subtraction | * Multiplication | / Division
- %/% Integer Division | %% Modulus (Remainder) | ^ Exponentiation

3. Comparison Operators:

- > Greater than | < Less than | == Equal to | != Not equal to | >= >= | <= <=

4. Logical Operators:

- & AND | | OR | ! NOT

5. Built-in Math Functions:

- sqrt(), abs(), round(), floor(), ceiling(), sum(), mean(), median(), var(), sd(), min(), max(), range()
- log(), log10(), log2(), exp() — Logarithmic and exponential functions.
- sin(), cos(), tan() — Trigonometric functions (input in radians).

Program 1a — Setting up Working Directory

Code for Program 1a — Working Directory Operations:

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1a_working_dir.R

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```r
# Experiment 1a: Setting up Working Directory in R

# Check current working directory
getwd()

# Set a new working directory
setwd("C:/Users/Student/R_Practicals")

# Verify the change
getwd()

# List all files in the working directory
list.files()

# List files with full path
list.files(full.names = TRUE)

# Display current working directory path
cat("Working Directory: ", getwd(), "\n")
```

Output for Program 1a:

Console ~/

```
> getwd()
[1] "C:/Users/Student/Documents"
> setwd("C:/Users/Student/R_Practicals")
> getwd()
[1] "C:/Users/Student/R_Practicals"
> list.files()
[1] "students.csv"  "students.txt"  "grade_A_students.csv"
> cat("Working Directory: ", getwd(), "\n")
Working Directory:  C:/Users/Student/R_Practicals
```

Program 1b — Basic Arithmetic Operations

Code for Program 1b — Arithmetic Operations:

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1b_arithmetic.R

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```r
# Experiment 1b: Basic Arithmetic Operations in R

a <- 25
b <- 7

cat("--- Arithmetic Operations ---\n")
cat("a =", a, "  b =", b, "\n")

cat("Addition       : a + b =", a + b, "\n")
cat("Subtraction    : a - b =", a - b, "\n")
cat("Multiplication : a * b =", a * b, "\n")
cat("Division       : a / b =", a / b, "\n")
cat("Integer Div    : a %/% b =", a %/% b, "\n")
cat("Modulus        : a %% b =", a %% b, "\n")
cat("Exponent       : a ^ 2  =", a ^ 2, "\n")

# Square root and absolute value
cat("Square Root    : sqrt(a) =", sqrt(a), "\n")
cat("Absolute Value : abs(-a) =", abs(-a), "\n")

# Rounding functions
x <- 3.7856
cat("Round 2 dec   : round(x,2) =", round(x, 2), "\n")
cat("Floor          : floor(x)  =", floor(x),"\n")
cat("Ceiling        : ceiling(x)=", ceiling(x), "\n")
```

Output for Program 1b:

Console ~/

```
> cat("--- Arithmetic Operations ---\n")
--- Arithmetic Operations ---
> cat("a =", a, "  b =", b, "\n")
a = 25   b = 7

Addition       : a + b = 32
Subtraction    : a - b = 18
Multiplication : a * b = 175
Division       : a / b = 3.571429
Integer Div    : a %/% b = 3
Modulus        : a %% b = 4
Exponent       : a ^ 2  = 625

Square Root    : sqrt(a) = 5
Absolute Value : abs(-a) = 25

Round 2 dec   : round(x,2) = 3.79
Floor          : floor(x)   = 3
Ceiling        : ceiling(x) = 4
```

Program 1c — Comparison and Logical Operators

Code for Program 1c — Comparison and Logical Operators:

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1c_operators.R
Source on Save ▶ Run Source ▾

```
# Experiment 1c: Comparison and Logical Operators

p <- 15
q <- 9

cat("--- Comparison Operators ---\n")
cat("p > q  :", p > q,  "\n")
cat("p < q  :", p < q,  "\n")
cat("p == q :", p == q, "\n")
cat("p != q :", p != q, "\n")
cat("p >= q :", p >= q, "\n")
cat("p <= q :", p <= q, "\n")

cat("--- Logical Operators ---\n")
m <- TRUE
n <- FALSE
cat("m AND n : m & n  =", m & n,  "\n")
cat("m OR  n : m | n  =", m | n,  "\n")
cat("NOT m   : !m     =", !m,     "\n")
cat("NOT n   : !n     =", !n,     "\n")
```

Output for Program 1c:

```
Console ~/
> cat("--- Comparison Operators ---\n")
--- Comparison Operators ---
p > q  : TRUE
p < q  : FALSE
p == q : FALSE
p != q : TRUE
p >= q : TRUE
p <= q : FALSE

> cat("--- Logical Operators ---\n")
--- Logical Operators ---
m AND n : m & n  = FALSE
m OR  n : m | n  = TRUE
NOT m   : !m     = FALSE
NOT n   : !n     = TRUE
```

Program 1d — Variables and Data Types

Code for Program 1d — Variables and Data Types:

```
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1d_variables.R
Source on Save    ▶ Run  Source ▾

# Experiment 1d: Variables and Data Types

# Numeric
num_val <- 98.6
cat("Numeric Value :", num_val, "| Class:", class(num_val), "\n")

# Integer
int_val <- 42L
cat("Integer Value :", int_val, "| Class:", class(int_val), "\n")

# Character
str_val <- "YCCE Nagpur"
cat("String Value  :", str_val, "| Class:", class(str_val), "\n")

# Logical
bool_val <- TRUE
cat("Logical Value :", bool_val, "| Class:", class(bool_val), "\n")

# Complex
cplx_val <- 3+4i
cat("Complex Value :", as.character(cplx_val), "| Class:", class(cplx_val), "\n")

# Type checking
cat("\n--- Type Checks ---\n")
cat("is.numeric(num_val)  :", is.numeric(num_val), "\n")
cat("is.integer(int_val)  :", is.integer(int_val), "\n")
cat("is.character(str_val):", is.character(str_val), "\n")
cat("is.logical(bool_val) :", is.logical(bool_val), "\n")
```

Output for Program 1d:

Console ~/

```
> cat("Numeric Value :", num_val, "| Class:", class(num_val), "\n")
Numeric Value : 98.6 | Class: numeric
> cat("Integer Value :", int_val, "| Class:", class(int_val), "\n")
Integer Value : 42 | Class: integer
> cat("String Value  :", str_val, "| Class:", class(str_val), "\n")
String Value  : YCCE Nagpur | Class: character
> cat("Logical Value :", bool_val, "| Class:", class(bool_val), "\n")
Logical Value : TRUE | Class: logical
> cat("Complex Value :", as.character(cplx_val), "| Class:", class(cplx_val), "\n")
Complex Value : 3+4i | Class: complex

--- Type Checks ---
is.numeric(num_val)  : TRUE
is.integer(int_val)  : TRUE
is.character(str_val): TRUE
is.logical(bool_val) : TRUE
```

Program 1e — Built-in Mathematical Functions

Code for Program 1e — Mathematical Functions:

R RStudio

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1e_math_functions.R

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```r
# Experiment 1e: Built-in Mathematical Functions

vals <- c(4, 9, 16, 25, 36)

cat("--- Math Functions on Vector ---\n")
cat("Values     :", vals, "\n")
cat("Sum        :", sum(vals), "\n")
cat("Mean       :", mean(vals), "\n")
cat("Median     :", median(vals), "\n")
cat("Variance   :", var(vals), "\n")
cat("Std Dev    :", sd(vals),  "\n")
cat("Min        :", min(vals), "\n")
cat("Max        :", max(vals), "\n")
cat("Range      :", range(vals), "\n")
cat("Sqrt each  :", sqrt(vals), "\n")

# Logarithm and exponential
cat("\n--- Log & Exp ---\n")
cat("log(100)   :", log(100), "\n")
cat("log10(100) :", log10(100), "\n")
cat("log2(8)    :", log2(8),  "\n")
cat("exp(1)     :", exp(1),  "\n")

# Trigonometric functions (in radians)
cat("\n--- Trigonometry ---\n")
cat("sin(pi/2)  :", sin(pi/2), "\n")
cat("cos(pi)    :", cos(pi),   "\n")
cat("tan(pi/4)  :", tan(pi/4), "\n")
```

Output for Program 1e:

```
Console ~/
> cat("--- Math Functions on Vector ---\n")
--- Math Functions on Vector ---
Values     : 4  9  16  25  36
Sum        : 90
Mean       : 18
Median     : 16
Variance   : 166
Std Dev    : 12.88410
Min        : 4
Max        : 36
Range      : 4  36
Sqrt each  : 2  3  4  5  6

--- Log & Exp ---
log(100)   : 4.605170
log10(100) : 2
log2(8)    : 3
exp(1)     : 2.718282

--- Trigonometry ---
sin(pi/2)  : 1
cos(pi)    : -1
tan(pi/4)  : 1
```

Conclusion

Thus, the following R programs for Experiment 1 were successfully written, executed in RStudio, and the outputs were verified:

- Program 1a: The working directory was set using setwd(), verified using getwd(), and all files in the directory were listed using list.files(). This establishes the correct file path context for all subsequent R programs.
- Program 1b: Basic arithmetic operations were performed on variables a=25 and b=7. All operators — addition (+), subtraction (-), multiplication (*), division (/), integer division (%/%), modulus (%%), and exponentiation (^) — were demonstrated along with rounding functions round(), floor(), and ceiling().
- Program 1c: Comparison operators (>, <, ==, !=, >=, <=) and logical operators (&, |, !) were applied to numeric and logical variables. All operators produced correct Boolean (TRUE/FALSE) results as expected.
- Program 1d: Five R data types — numeric, integer, character, logical, and complex — were declared and printed. The class() and is.*() functions were used to verify the type of each variable, confirming R's dynamic type system.
- Program 1e: Built-in mathematical functions including sum(), mean(), median(), var(), sd(), min(), max(), range(), sqrt(), log(), log10(), log2(), exp(), sin(), cos(), and tan() were applied to verify R's comprehensive math library.

All programs were executed in RStudio (R v4.5.3) and produced the expected outputs. The working directory setup and basic calculation capabilities of R were successfully demonstrated.